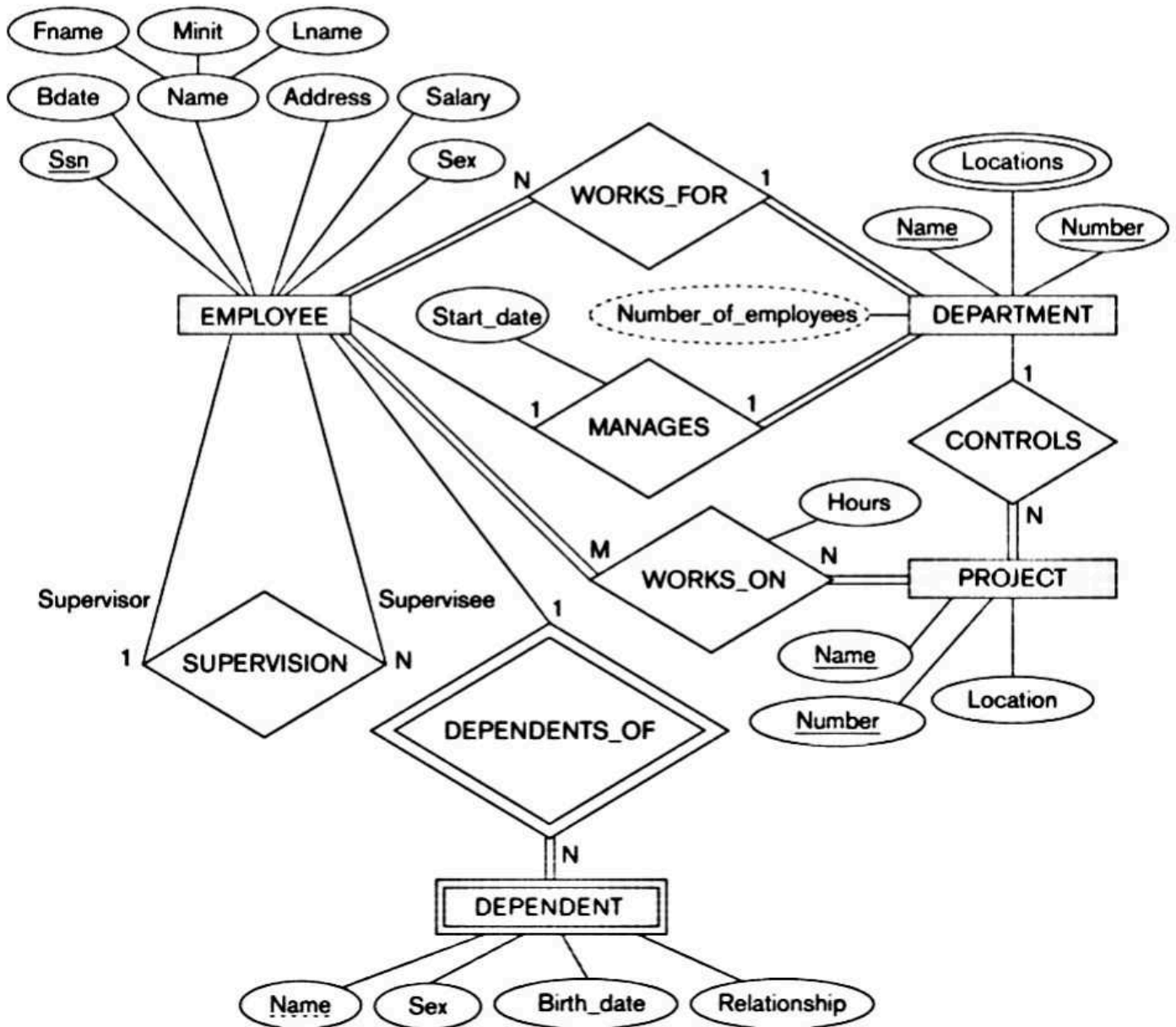


LATIHAN 1

Mapping dari ER-Diagram ke Relational Schema

Diketahui ER-Diagram sebagai berikut:



Berdasarkan gambar ER-Diagram diatas:

1. Jelaskan masing-masing Entitas dan Atribut
2. Jelaskan simbol-simbol yang digunakan
3. Jelaskan relasi antar masing-masing entitas dan derajat relasinya
4. Jelaskan terkait total participation dan partial participation
5. Lakukan proses mapping ER-Diagram menjadi tabel-tabel atau Relational Schema

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1. Entities and Attributes

- Employee is a strong entities that have attributes like Ssn (Primary key), Name (Composite Attribute), Consisting of Fname (First name), Minit (Middle Initial), Lname (Last name), Bdate (birth date), Address, Salary, Sex or gender
- Departement is a strong entities that have attributes like Number (Primary key), Name, locations, Number of employees (derived attribute) calculated from employee data
- Project is a strong entities that have attributes like Number (Primary key), Name, Location
- Dependent is a weak entities that have attributes like Name (Partial key), Sex, Birth - date, Relationship

2. Symbols used in the ER Diagram

- Rectangle Represents Entity
- Oval / Ellipse represents Attribute
- Double Oval represents multivalued attribute
- Dashed oval represents Derived attribute
- Diamond represents Relationship
- Double Rectangle Represents weak entity
- Single line represents Partial Participation
- Double line represents Total Participation
- Numbers (1, N, M) represents Cardinality of the relationship

3. Relationship between entities and their degree

- relationship works - for with cardinality N:1 means many employees work for one department
- relationship manages with cardinality 1:1 means one employee manages one department
- relationship Controls with cardinality 1:N means One

department controls multiple projects

- relationship works on with cardinality $M:N$ means An employee can work on multiple projects and a project can have many employees

- relationship Supervision with cardinality $1:N$ means one employee (as supervisor) supervises multiple employees (as supervised)

- relationship Dependents - of with cardinality $1:N$ means one employee can have multiple dependents

4. Total Participation and Partial Participation

- Total Participation occurs when every entity must participate in a relationship

example: Dependent has total participation in Dependents
- OF, because a dependent must belong to an employee

- Partial Participation occurs when not all entities must participate in a relationship

example: Employee in Manages, because not all employees are managers.

5. Mapping ER Diagram to Relational Schema

Employee									
<u>Fname</u>	Minit	Lname	<u>SSN</u>	Bdate	Address	Sex	Salary	SuperSSN	DNO

Department			
<u>DName</u>	DNumber	MGRSSN	MGRSTARTDATE

Dept - Locations	
DNumber	<u>DLocation</u>

Project			
<u>PName</u>	<u>PNumber</u>	PLocation	<u>DNum</u>

Works on		
<u>E SSN</u>	<u>PNO</u>	HOURS

Dependent				
<u>E SSN</u>	<u>Dependent name</u>	Sex	Bdate	Relationship